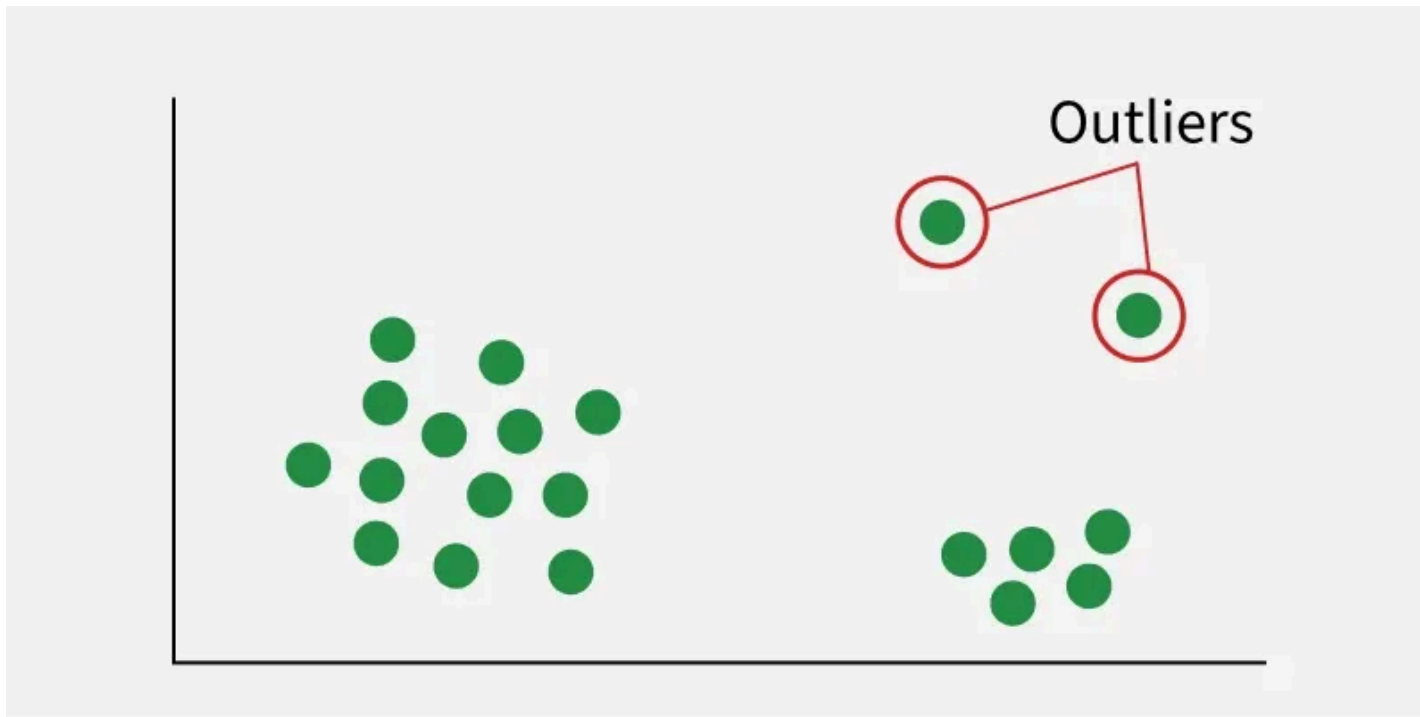


Outliers



Outliers are **observations (data points) that significantly differ from the majority of the data**. They may represent:

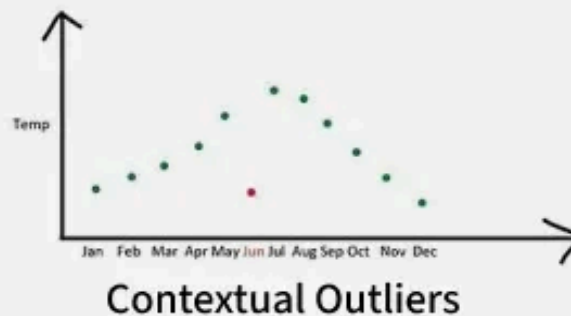
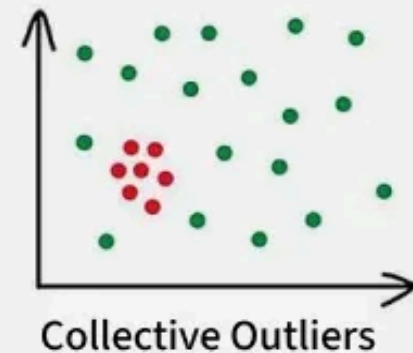
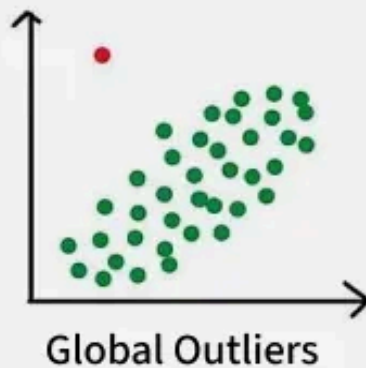
- Genuine rare events
- Measurement errors
- Fraudulent activities
- Novel or unexpected behavior

Outliers are mainly classified into **three types**:

1. **Global (Point) Outliers**
2. **Contextual (Conditional) Outliers**
3. **Collective Outliers**

Outliers are mainly classified into three types

Types of Outliers



1. Global Outlier (Point Outlier)

A **global outlier** is a data point that is extremely different from the rest of the dataset **without considering any context**.

It looks unusual when compared to the entire dataset.

Example

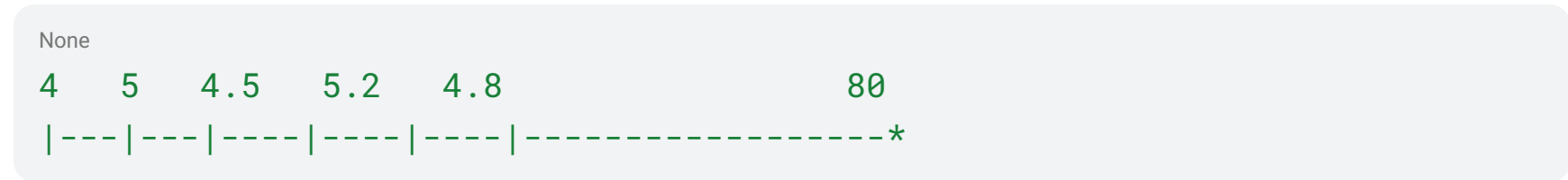
Suppose these are employee salaries (in ₹ Lakhs):

Employee	Salary
A	4
B	5
C	4.5
D	5.2
E	4.8
F	80

Here,

₹80 Lakhs is much higher than everyone else. It is a **global outlier**.

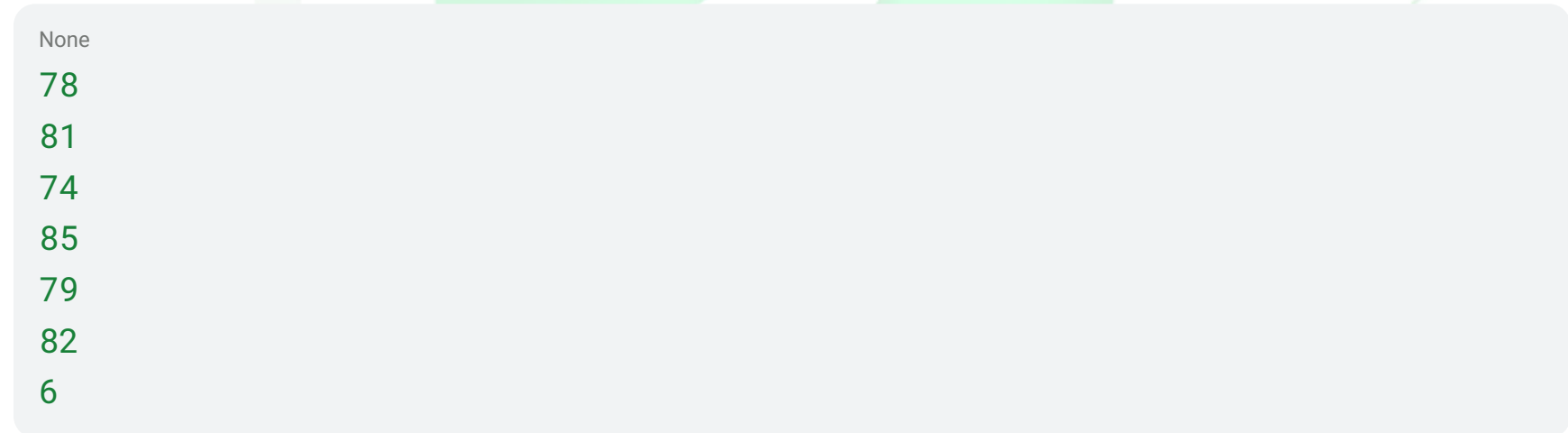
Visualization



The star is far away from the majority.

Another Example

Student Marks



Score **6** is clearly abnormal.

Real-Life Example

Credit card transactions

Typical purchases:

None

₹500

₹700

₹1200

₹900

₹650

One purchase:

None

₹12,00,000

This is globally abnormal.

Characteristics

- Easy to identify
- No additional information needed
- Compared against the whole dataset

2. Contextual Outlier (Conditional Outlier)

A **contextual outlier** is normal in one situation but abnormal in another.

Here, **context matters**.

The value itself may not be unusual.

Its relationship with another feature makes it abnormal.

Example 1: Temperature

Imagine these temperatures.

Month	Temperature
January	8°C
February	10°C
March	15°C
July	36°C
August	35°C

Everything looks fine.

Now suppose

Month	Temperature
January	36°C

36°C is not unusual overall.

But **36°C in January** is highly unusual.

This is a contextual outlier.

Why?

Without context:

None

36°C

is normal. With context:

None

Month = January

Temperature = 36°C

it becomes abnormal.

Example 2: Electricity Consumption

Normal daily usage:

None

Morning : 4 units

Afternoon : 6 units

Evening : 8 units

Now

None

2 AM : 8 units

8 units isn't inherently abnormal.

But at **2 AM**, it is unusual.

Example 3: Bank Transactions

Someone usually spends

None

₹40,000 during vacation

Perfectly normal. But

None

₹40,000 at 3:00 AM in another country

This becomes suspicious.

Context Features

Examples include:

- Time
- Location
- Season
- Weather
- User
- Age
- Device
- Day of the week

Characteristics

- Needs additional variables (context)
- Very common in time-series
- Harder than global outlier detection

3. Collective Outlier

A **collective outlier** occurs when **a group of observations is abnormal together**, even though each individual observation may appear normal.

Each point alone is fine.

Together they form an unusual pattern.

Example 1: ECG Signal

Normal heartbeat:

None



Abnormal heartbeat:

None



Each heartbeat looks normal.

But the sequence is abnormal.

This is a collective outlier.

Example 2: Cybersecurity

Login attempts:

None

```
10:00 Login
10:05 Logout
11:00 Login
11:20 Logout
```

Normal.

Now imagine:

None

```
10:00 Login
10:01 Login
10:02 Login
10:03 Login
10:04 Login
10:05 Login
```

Each login is valid.

The pattern is suspicious.

Example 3: Credit Card Fraud

Purchases:

None

₹250

₹180

₹320

₹275

₹190

Each purchase is normal.

But:

None

100 purchases within 2 minutes

The sequence is abnormal.

Example 4: Sensor Data

Machine vibration:

None

2.1

2.0

2.2

2.1

2.0

Normal.

Now:

None

2.0

2.1

2.2

2.0

2.1

2.2

2.0

2.1

2.2

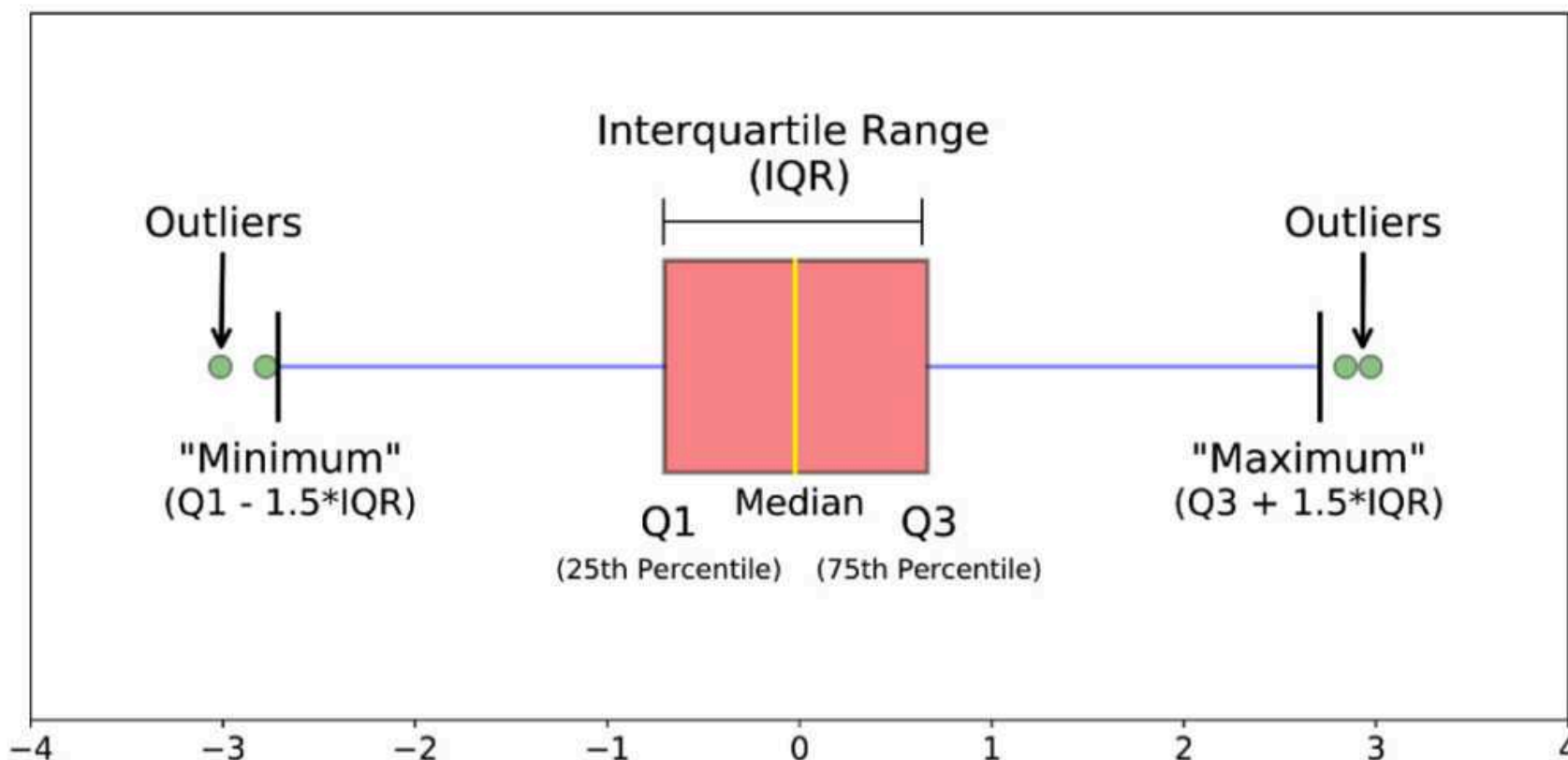
Each value is normal.

But the repeated oscillating pattern may indicate a fault.

Characteristics

- Individual observations look normal
- The sequence or group is abnormal
- Common in:
 - Fraud detection
 - Network intrusion
 - Medical monitoring
 - Sensor analysis
 - Video surveillance

Box Plot



A **box plot (box-and-whisker plot)** is one of the most powerful tools in **Exploratory Data Analysis (EDA)**. It summarizes the distribution of data using the **five-number summary** and helps identify **outliers, skewness, spread, and central tendency**.

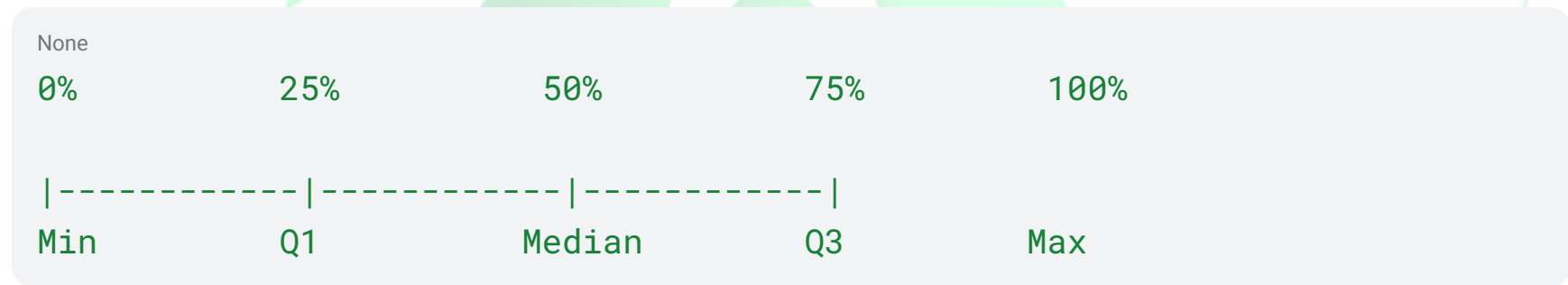
Step 1: Find the **Median** (Q2)

Step 2: Find Q1 : Q1 is the **median of the lower half**.

Step 3: Find Q3 : Q3 is the **median of the Upper half**.

Understanding Quartiles

Quartiles divide the data into **four equal parts**.



Meaning:

- 25% of observations are below Q1.
- 50% are below the median.
- 75% are below Q3.
- 25% lie above Q3.

What is Q1?

Q1 is the **25th percentile**.

It means:

25% of the observations are smaller than Q1.

Example:

None

2 4 5 7 | 8 | 9 10 12 15

Q1 lies around **4.5**.

What is Q3?

Q3 is the **75th percentile**.

It means:

75% of observations are below Q3.

What is the IQR (Interquartile Range)?

The **Interquartile Range (IQR)** measures the spread of the **middle 50%** of the data.

Formula:

$$\text{IQR} = Q3 - Q1$$

For our example:

$$\text{IQR} = 11 - 4.5 = 6.5$$

The middle 50% of the data spans **6.5 units**.

Why use the IQR?

The IQR is **robust** because it ignores extreme values.

Example:

None

2 4 5 7 8 9 10 12 15 1000

The value **1000** dramatically increases the range but does not affect Q1 or Q3 much, so the IQR remains a reliable measure of spread.

What are Fences?

Fences are boundaries used to detect **potential outliers**.

There are two fences:

- Lower Fence
- Upper Fence

Formulas:

Lower Fence = $Q1 - 1.5 \times IQR$

Upper Fence = $Q3 + 1.5 \times IQR$

Why multiply by 1.5?

The factor **$1.5 \times IQR$** is a widely used rule introduced by John Tukey. It identifies observations that are unusually far from the central 50% of the data without assuming the data follows a normal distribution.

So:

- Lower Fence = **-5.25**
- Upper Fence = **20.75**

Detecting Outliers

Any observation:

- Less than **-5.25**
- Greater than **20.75**

is considered a **potential outlier**.

Suppose the data becomes:

None

2 4 5 7 8 9 10 12 15 45

Since:

None

45 > 20.75

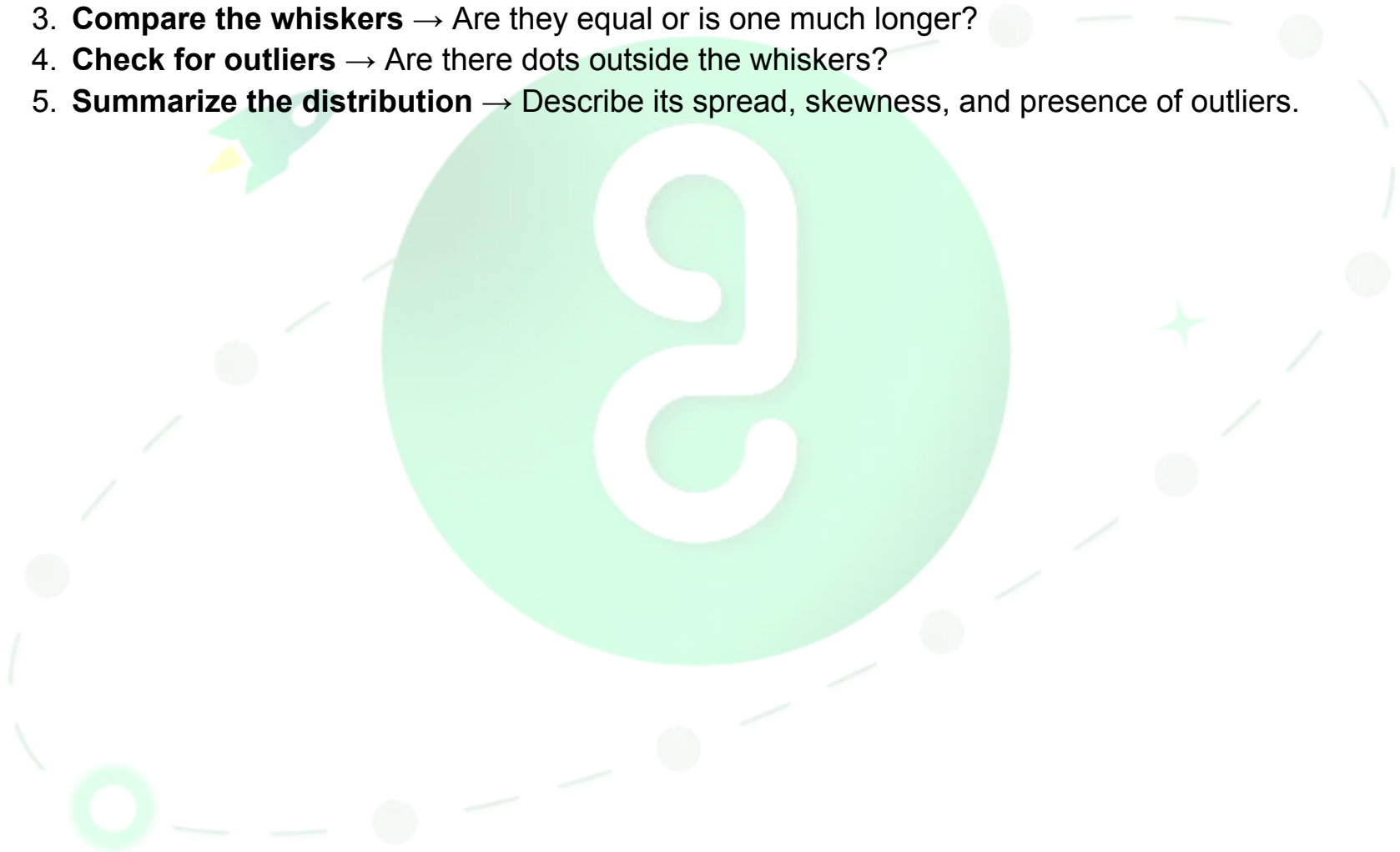
45 is a potential outlier.

Quick Interpretation Cheat Sheet

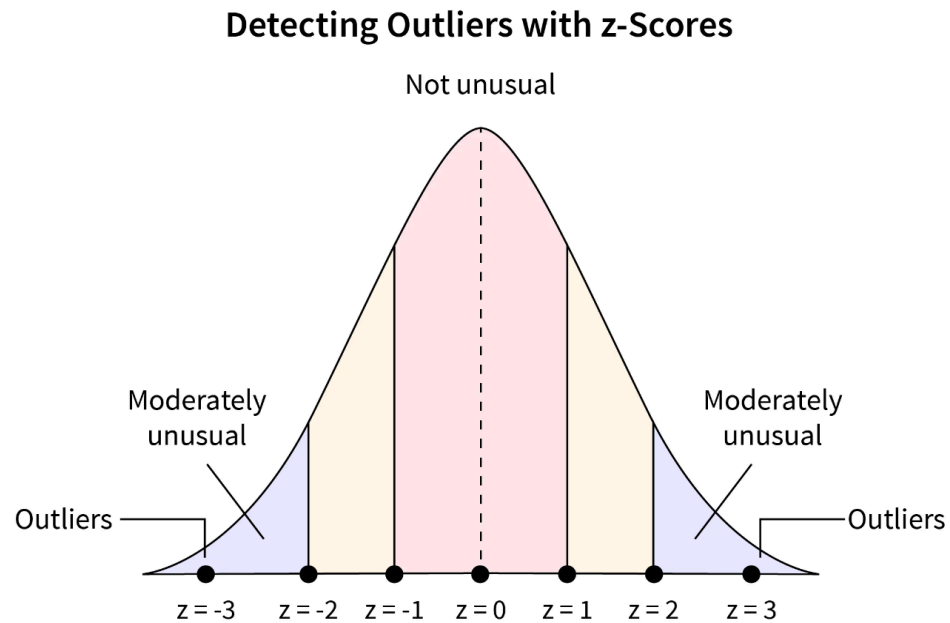
What You See	What It Means
Small box (small IQR)	Low variability; data is tightly clustered
Large box (large IQR)	High variability; data is widely spread
Median in the center	Approximately symmetric distribution
Median near Q1	Right (positive) skew; upper half is more spread out
Median near Q3	Left (negative) skew; lower half is more spread out
Equal whiskers	Balanced spread on both sides
Long upper whisker	Long upper tail; right-skewed distribution
Long lower whisker	Long lower tail; left-skewed distribution
One or more dots beyond whiskers	Potential outliers that should be investigated
No outliers	No observations beyond the $1.5 \times \text{IQR}$ fences

A simple 5-step approach to interpreting any box plot

1. **Look at the box** → Is the spread (IQR) small or large?
2. **Look at the median** → Is it centered or closer to Q1/Q3?
3. **Compare the whiskers** → Are they equal or is one much longer?
4. **Check for outliers** → Are there dots outside the whiskers?
5. **Summarize the distribution** → Describe its spread, skewness, and presence of outliers.



Z-score method



The **Z-score method** is one of the most popular statistical techniques for detecting outliers. It measures **how many standard deviations a data point is away from the mean**.

It is most effective when the data is **approximately normally distributed**.

What is a Z-score?

A **Z-score (Standard Score)** tells us the position of a data point relative to the mean.

- Positive Z-score → Above the mean
- Negative Z-score → Below the mean
- Zero → Exactly at the mean

Mathematically

$$Z = \frac{x - \mu}{\sigma}$$

Where:

- x = Data point
- μ = Mean
- σ = Standard deviation

Why do we use Z-score?

Suppose two students score:

Student	Marks
A	80
B	80

Do they perform equally? Not necessarily.

Class A

- Average = 78
- Standard deviation = 2

Class B

- Average = 50
- Standard deviation = 20

Although both scored **80**, Student A is close to the class average, while Student B is far above the average.

The Z-score quantifies this difference.

Understanding the Formula

Suppose:

Mean = **50**

Standard deviation = **10**

Find the Z-score for **70**.

Step 1:

$$x - \mu = 70 - 50 = 20$$

Step 2:

Divide by the standard deviation.

$$Z = 20 / 10 = 2$$

Interpretation:

70 is two standard deviations above the mean.

What does a Z-score mean?

Z-score	Interpretation
0	Exactly at the mean
+1	One standard deviation above the mean
-1	One standard deviation below the mean
+2	Two standard deviations above the mean
-2	Two standard deviations below the mean
+3	Three standard deviations above the mean
-3	Three standard deviations below the mean

Understanding the ± 3 Rule

For normally distributed data, almost all observations lie close to the mean.

This is summarized by the **68–95–99.7 Rule**.

Range	Percentage of Data
Mean $\pm 1\sigma$	~68%
Mean $\pm 2\sigma$	~95%
Mean $\pm 3\sigma$	~99.7%

Only about **0.3%** of observations fall beyond ± 3 standard deviations.

That is why values with:

$Z < -3$

or

$Z > +3$

are considered **potential outliers**.

Visual detection: scatter plot, histogram

Comparison

Scatter Plot	Histogram
Shows individual observations	Groups observations into bins
Easy to identify isolated points	Easy to understand overall distribution
Good for relationships between variables	Good for frequency distribution
Best for detecting isolated outliers	Best for identifying skewness and gaps

Which Visual Method Should You Use?

Scenario	Recommended Plot
Detect individual outliers	Scatter Plot
Understand data distribution	Histogram
Check skewness	Histogram + KDE
Explore relationships between two variables	Scatter Plot
Confirm statistical outliers	Box Plot + IQR/Z-score

Comparison of Treatment Methods

Method	What it Does	Data Loss	Best For
Remove	Deletes outliers	Yes	Errors, invalid data, duplicate records
Winsorization (Cap)	Replaces outliers with lower/upper limits	No	Financial data, regression, preserving sample size
Log Transformation	Compresses large values	No	Right-skewed positive data (income, sales, prices)

Which Method Should You Use?

Situation	Recommended Method
Incorrect data entry (e.g., age = 500)	Remove
Genuine but extreme values	Winsorization

Highly right-skewed data	Log Transformation
Small dataset	Prefer capping over removing
Large dataset with erroneous values	Remove
Regression with influential outliers	Winsorization or Log Transformation